DCTCP Metrics Analysis

DCTCP (Data Center TCP):

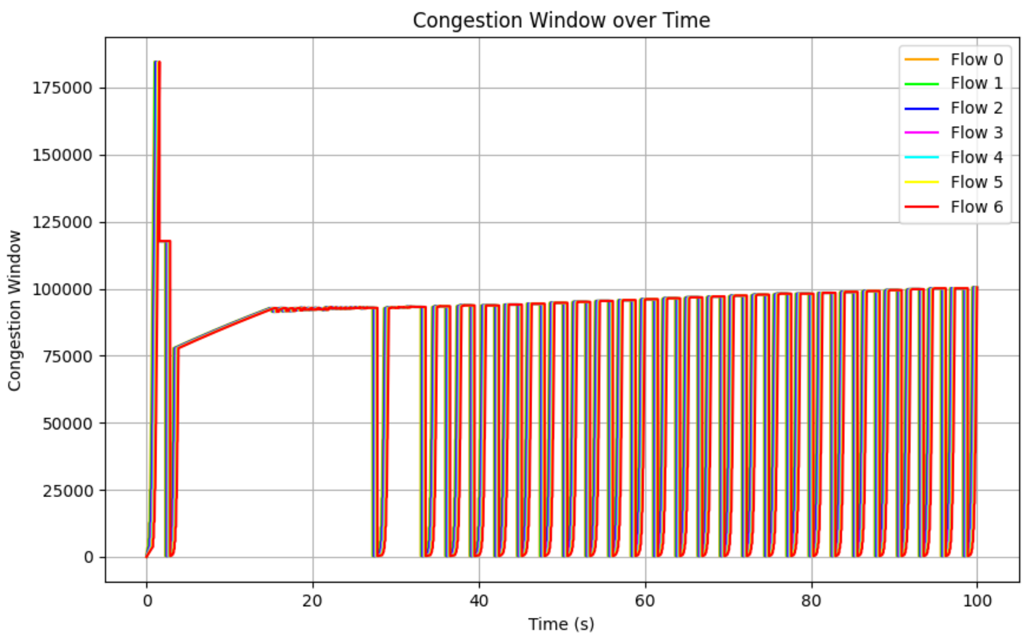
- **Purpose:** Designed for data center networks to achieve low latency and high throughput.
- **Key Feature:** Uses Explicit Congestion Notification (ECN) to provide fine-grained congestion feedback.

How It Works: DCTCP uses ECN to detect and respond to congestion without relying on packet loss. Switches mark packets with ECN when queue lengths exceed a threshold, and the receiver informs the sender about these marks. The sender adjusts its congestion window proportionally to the fraction of marked packets, ensuring low latency and high throughput in data center environments.

- **Congestion Estimate (α):** The algorithm maintains a moving average of congestion, represented by the α parameter. This value is updated during each observation window based on ECN-marked and total acknowledged bytes.
- **ECN Handling:** DCTCP requires ECN capabilities at both sender and receiver. It uses either ECT(0) or ECT(1) codepoints to identify congestion marks.
- **State Transitions:** The implementation handles transitions between congestion states (e.g., CE state) and updates internal parameters accordingly.
- **Observation Windows:** The congestion estimate is recalculated for every observation window, resetting metrics like total acknowledged bytes and ECN-marked bytes.

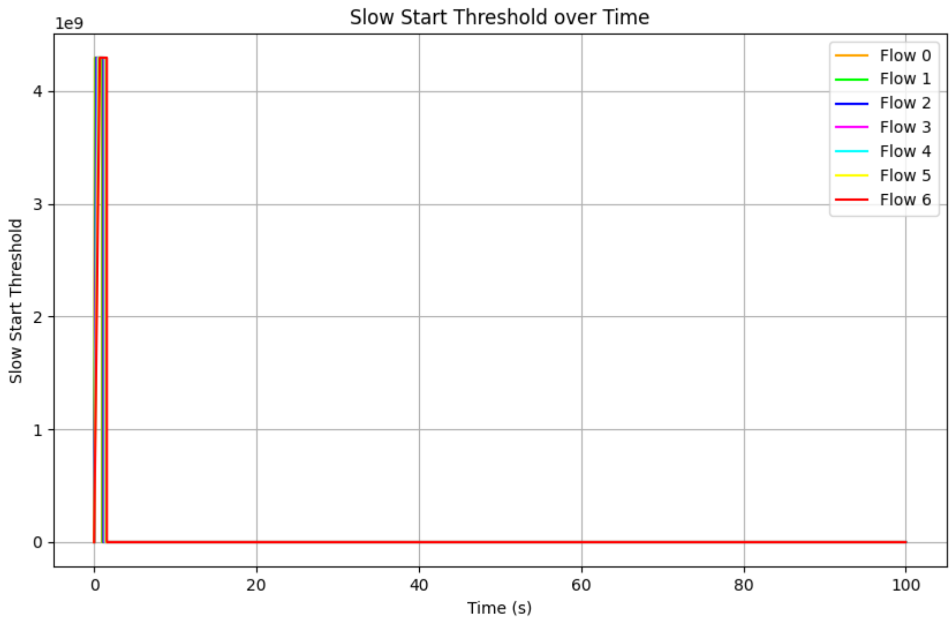
Graphs

Congestion Window (cwnd): A TCP state variable that controls the amount of data a TCP sender can send into the network before receiving an acknowledgment (ACK) for previously sent data.



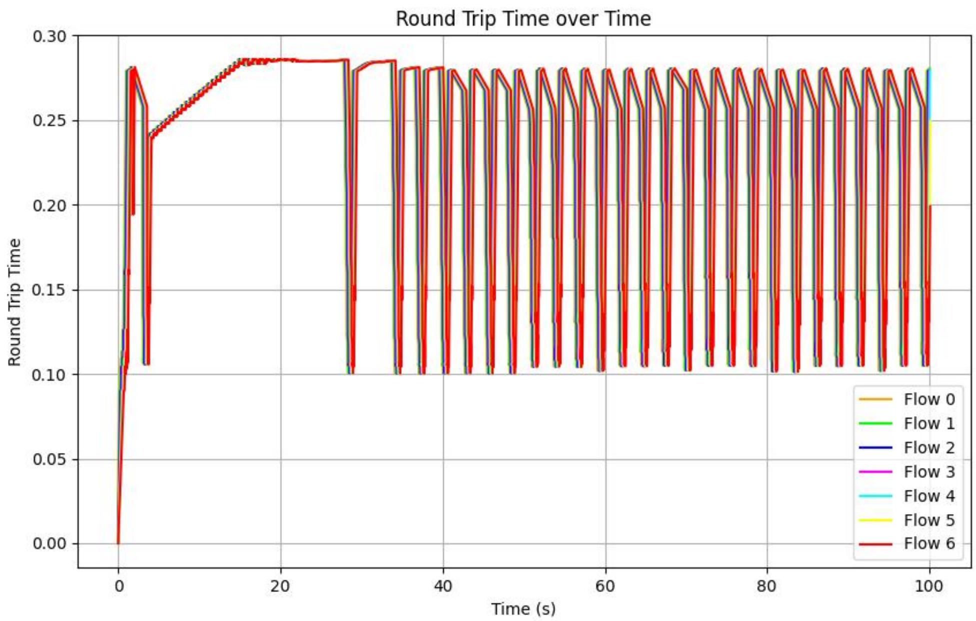
- Initial exponential growth (slow start phase) to quickly utilize bandwidth.
- Proportional reduction in CWND in response to ECN marks, avoiding drastic drops.
- Stabilization of CWND, with periodic oscillations reflecting the network's congestion level.
- When ECN mark is triggered the CWND is reset again and gradually increases the mark

Slow Start Threshold(ssthresh): The slow start threshold determines when TCP switches from aggressive exponential growth (slow start phase) to more cautious linear growth (congestion avoidance phase). It's like a speed limit that TCP sets for itself based on previous congestion experiences. When cwnd reaches ssthresh, TCP becomes more conservative in its growth.



- In DCTCP, the **Slow Start Phase** does not play a significant role after the initial connection establishment.
- DCTCP primarily relies on **ECN (Explicit Congestion Notification)** and the **alpha parameter** to manage congestion.
- The SSTH is no longer needed because DCTCP avoids drastic window reductions (like halving CWND in traditional TCP) and instead reduces CWND smoothly based on congestion signals.

Round Trip Time (RTT): RTT measures how long it takes for a packet to reach its destination and for an acknowledgment to return.

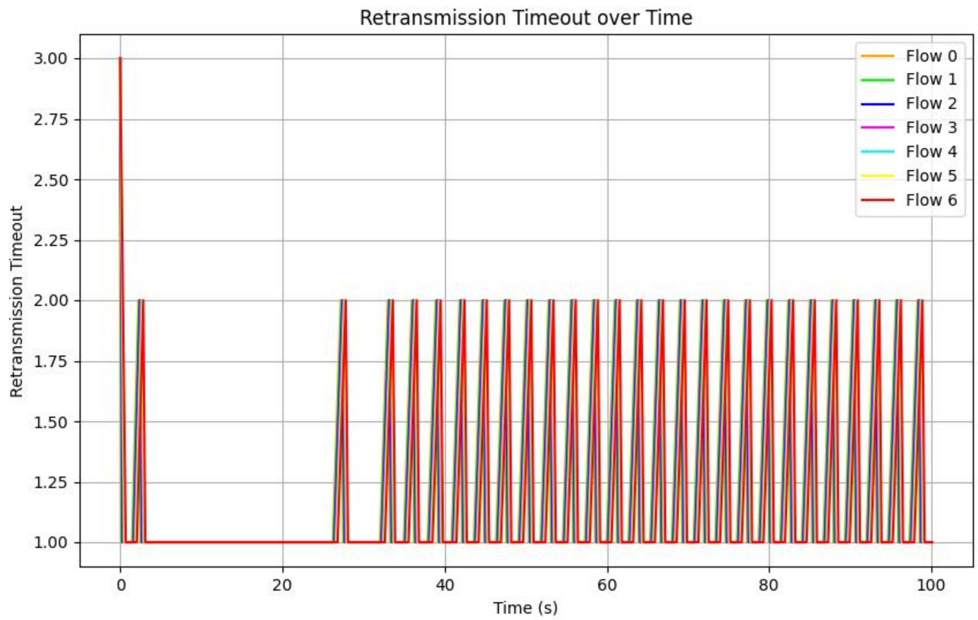


• The initial increase in RTT reflects the buildup of in-flight packets during the slow start phase as the congestion window grows exponentially.

• The periodic dips and recoveries in RTT correspond to DCTCP's proactive adjustments based on ECN signals, which prevent excessive queuing delays.

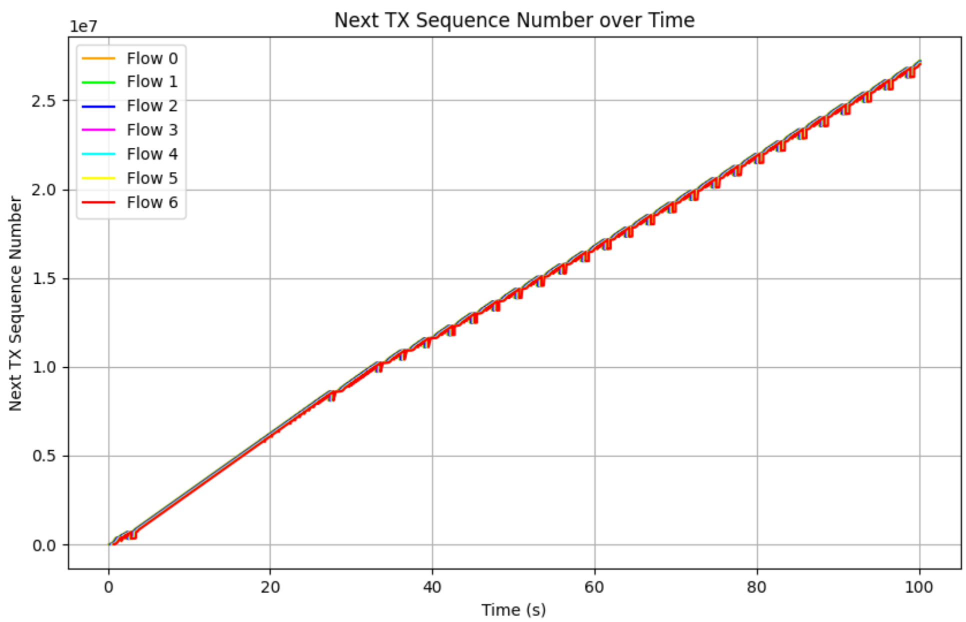
• The steady oscillations after the initial phase indicate a stable equilibrium maintained by DCTCP, balancing throughput and minimizing latency.

Retransmission Timeout (RTO): RTO determines how long TCP waits before assuming a packet is lost and needs retransmission. It's like a safety timer - if no acknowledgment arrives within this time, TCP assumes the worst and retransmits. The RTO value is typically calculated based on RTT measurements and their variance.



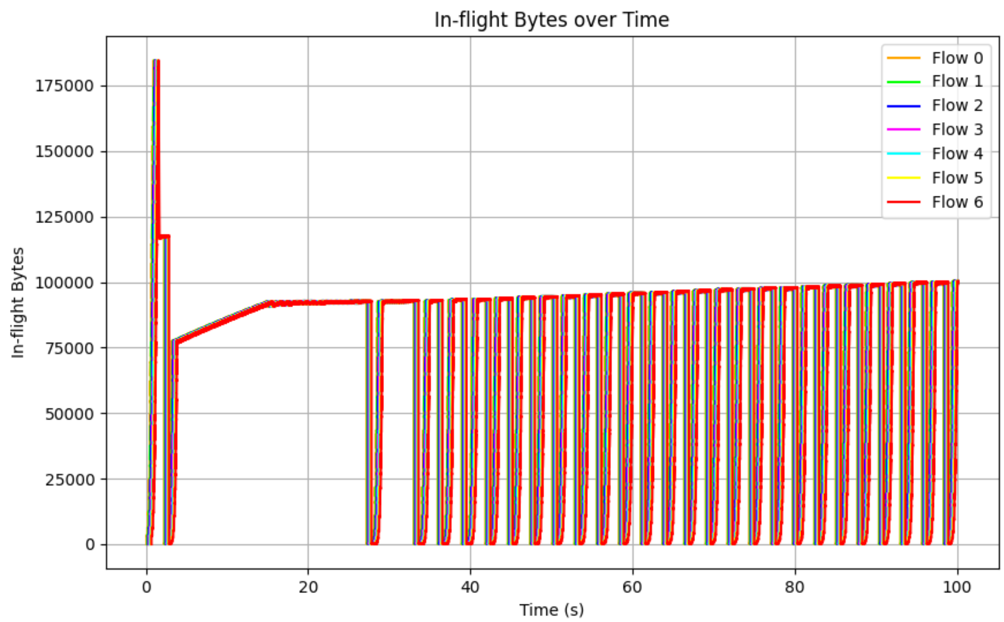
- RTT measurements remain stable, leading to a consistent and low RTO value.
- During (5 - 30) phase, the network experiences little or no congestion.
- The RTO is recalculated based on the new RTT value, resulting in a spike in RTO.

Next Transmission Sequence Number (next-tx): This tracks the sequence number of the next packet to be transmitted. It's like a counter that helps ensure packets are sent in order.



- When CWND shrinks (due to congestion), the sender reduces its sending rate, causing temporary **steps** in the Next TX graph.
- When CWND grows, the sender transmits data at a faster rate, causing the **Next TX sequence number** to increase linearly.

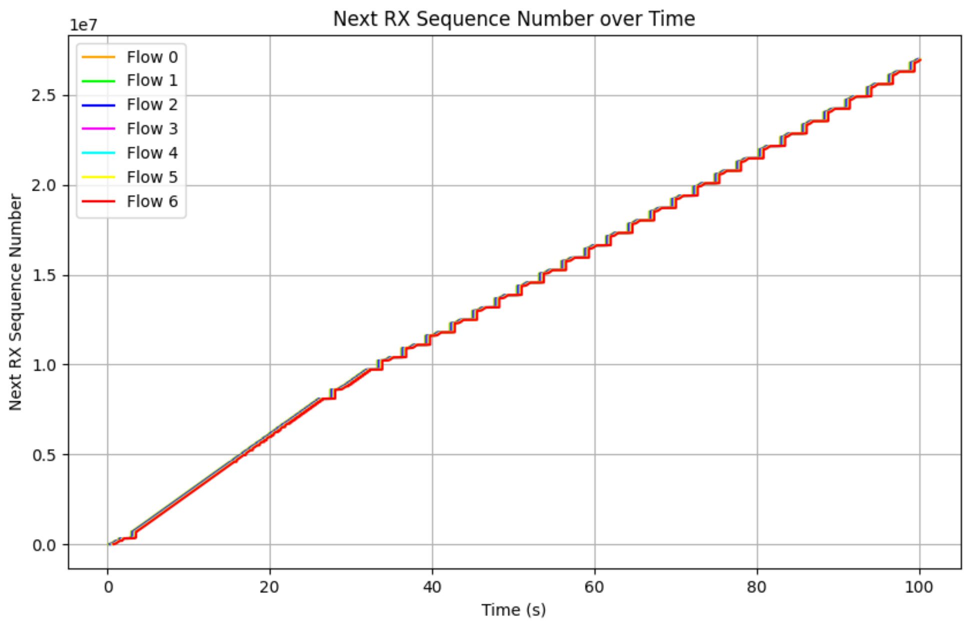
In-flight Bytes: This measures how many bytes are currently in transit - sent but not yet acknowledged. It's like tracking packages that are currently in delivery.



CWND Controls the Inflight Packets:

- CWND determines the maximum number of unacknowledged packets a sender can have in the network at any time.
- If CWND is large, the sender can have more packets in flight.
- If CWND is reduced (e.g., due to ECN marks), the inflight packets also decrease.

Next Receive Sequence Number (next RX): This indicates which sequence number the receiver expects next. It helps detect missing or out-of-order packets.



This graph completely depends on the (Tx vs T) graphs.